

A Native Reset Support Floor

Sharp Endpoint Composition and a 62-Step Common Finite Tube

Prime Dynamics Theory Program

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Abstract

The reset route now has three compatible ingredients. RH-153 transports a correlated Gram–tail pair through an overlap congruence without changing its relative tail. RH-154 supplies a well-conditioned terminal half. RH-155 constructs a native recent/tail pair with a subunit Loewner ratio at all 130 frozen snapshots.

We compose these ingredients in one sharp endpoint theorem. Suppose the selected full-memory eigenvalues lie in $[\ell, u]$, the old tail has operator mass at most τ , and the reset overlap singular values lie in $[\alpha, \beta]$. If $\ell > 2\tau$, then the transported native support satisfies

$$\mathcal{S} \geq \frac{\alpha}{\beta} \sqrt{\frac{\ell - \tau}{u}} \left(1 - \sqrt{\frac{\tau}{\ell - \tau}}\right)^4.$$

The formula is jointly sharp from these scalar endpoints.

All 120 frozen transitions have positive support lower. The minimum is 3.26204×10^{-8} , the median is 3.26385×10^{-4} , and 111 and 74 transitions exceed 10^{-6} and 10^{-4} respectively. All 62 terminal-half transitions form a common finite tube with the same minimum. This closes the finite native reset architecture. The support functional is not yet identified with the projected-cross directional quantity used in the earlier Stage-A route.

1 The endpoint data

Let $M = R + D$ be the full, recent, and old-tail memories compressed to the exact reset packet. RH-155 gives

$$\ell I \preceq M \preceq uI, \quad 0 \preceq D \preceq \tau I,$$

so

$$R \succeq (\ell - \tau)I, \quad \text{quad} D \preceq xR, \quad \text{quad} x = \frac{\tau}{\ell - \tau}.$$

Let C be an invertible overlap map with singular values in $[\alpha, \beta]$, and transport both matrices by $\Phi_C(A) = C^{-*}AC^{-1}$. The relative tail x is unchanged by RH-153.

For a positive pair define the native support functional

$$\mathcal{S}(G, D) = \sqrt{\frac{\lambda_{\min}(G)}{\lambda_{\max}(G)}} \left(1 - \sqrt{\gamma^2(G, D)}\right)_+^4,$$

where $\gamma^2(G, D) = \inf\{y : D \preceq yG\}$.

2 Sharp support composition

Theorem 1 (Native reset support theorem). *Assume $0 < \alpha \leq \beta$, $0 \leq \tau < \ell \leq u$, and use the endpoint data above. Then*

$$\mathcal{S}(\Phi_C(R), \Phi_C(D)) \geq \frac{\alpha}{\beta} \sqrt{\frac{\ell - \tau}{u}} \left(1 - \sqrt{\frac{\tau}{\ell - \tau}}\right)_+^4.$$

The right side is positive exactly when $\ell > 2\tau$ and $\alpha > 0$. The complete expression is sharp from the five scalar endpoints.

Proof. The recent eigenvalue endpoints give

$$a(R) := \sqrt{\lambda_{\min}(R)/\lambda_{\max}(R)} \geq \sqrt{(\ell - \tau)/u},$$

because $R \preceq M \preceq uI$. Inverse congruence lowers this normalized base by at most α/β . Loewner covariance preserves the tail ratio, which is at most $\tau/(\ell - \tau)$. The fourth-power factor is decreasing, so multiplication proves the formula.

For sharpness, take two dimensions with $M = \text{diag}(u, \ell)$, $D = \text{diag}(0, \tau)$, and $C = \text{diag}(\alpha, \beta)$. Assigning the u direction to α and the $\ell - \tau$ direction to β attains the base endpoint, while the second direction attains the tail ratio. Hence all factors are simultaneously attained [Bhatia, 1997, Horn and Johnson, 2013]. $\square$

Corollary 2 (Common finite tube). *For any finite reset family satisfying the theorem, the minimum of the displayed endpoint lowers is a valid common support floor. Congruence conditioning is paid once per independently reset snapshot, not multiplied through a recursive chain.*

The last sentence is essential. The atlas is reanchored at every snapshot; we do not multiply 120 support lowers or inverse-overlap factors.

3 Frozen support audit

For each RH-153 transition we use its robust overlap lower and selected full-memory eigenvalue upper. The lower selected eigenvalue is the minimum of the independently reconstructed RH-153 and RH-155 outward values. The geometric tail mass is taken from RH-155. Since overlaps of orthonormal frames have $\beta \leq 1$, we set $\beta = 1$.

Every one of the 120 transitions is positive. The smallest lower, 3.26204×10^{-8} , is the terminal $\sigma = 0.01$ left snapshot. There the recent-base lower is 4.2620×10^{-7} and the conservative tail factor is 0.0765374. Thus the weakest support is caused by both terminal spectral thinning and the largest certified tail ratio, not by an overlap birth spike.

The median support lower is 3.26385×10^{-4} and the maximum is 0.197087. All 120 exceed 10^{-8} , 111 exceed 10^{-6} , and 74 exceed 10^{-4} . The RH-154 terminal-half family contains 62 transitions; all are positive and their common floor is again 3.26204×10^{-8} . Delayed start improves conditioning while leaving the terminal support wall visible.

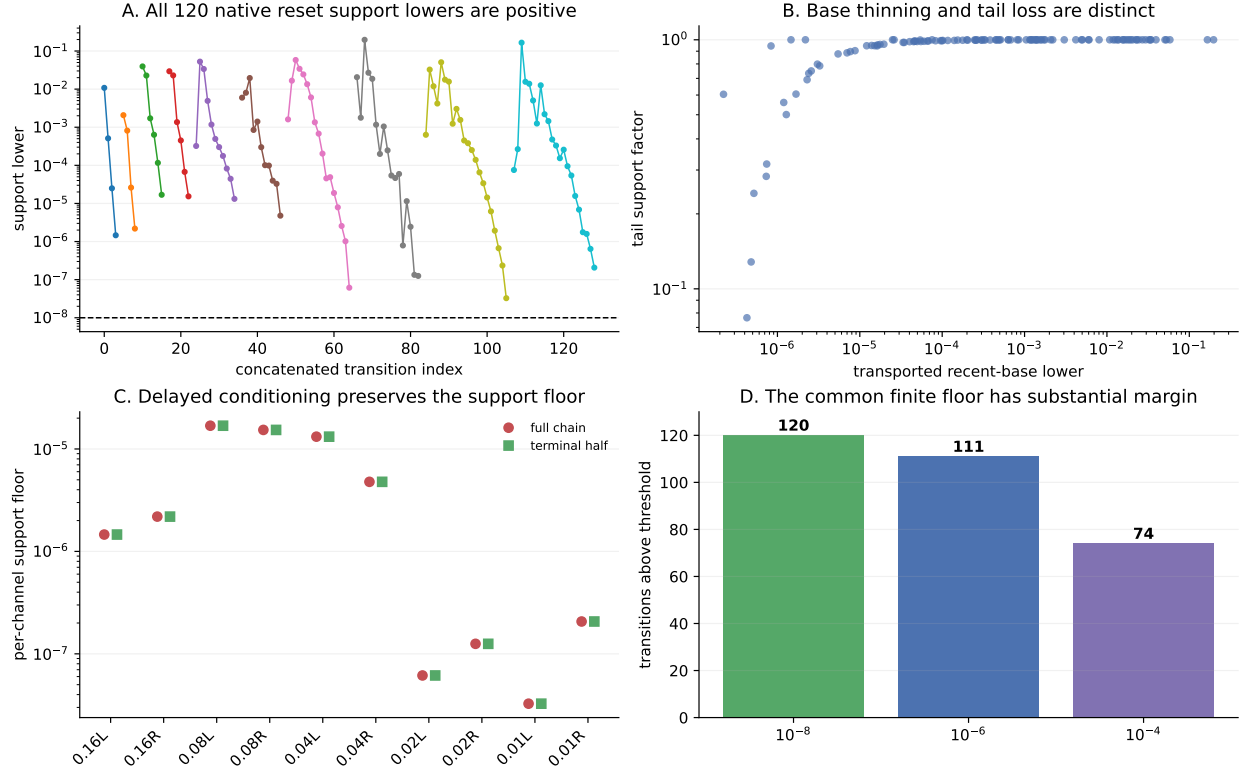


Figure 1: All transition support lowers, separation of base and tail losses, full-versus-delayed channel floors, and threshold counts.

4 A conditional all-level endpoint criterion

The finite formula also identifies the exact analytic packets needed for an eventual theorem. Let $(\ell_n, u_n, \tau_n, \alpha_n, \beta_n)$ be outward endpoint data for an all-level reset family and suppose each transition is reanchored independently.

Corollary 3 (Eventual endpoint criterion). *Assume that, after a finite index,*

$$\ell_n \geq \ell_* > 2\tau_*, \quad \tau_n \leq \tau_*, \quad u_n \leq u_* < \infty,$$

and

$$0 < \alpha_* \leq \alpha_n \leq \beta_n \leq \beta_* < \infty.$$

Then the native reset support has the uniform eventual floor

$$\liminf_{n \rightarrow \infty} \mathcal{S}_n \geq \frac{\alpha_*}{\beta_*} \sqrt{\frac{\ell_* - \tau_*}{u_*}} \left(1 - \sqrt{\frac{\tau_*}{\ell_* - \tau_*}} \right)^4 > 0.$$

For trace-one geometric memory one may take $\tau_* = \eta^d / (1 - \eta)$.

Proof. Apply Theorem 1 at every sufficiently large index and use monotonicity in the five endpoints. A finite omitted prefix does not alter the liminf. $\square$

Within this scalar endpoint architecture, none of the three qualitative requirements can simply be dropped. If $\alpha_n/\beta_n \rightarrow 0$, the chart can flatten the normalized base. If $\ell_n \downarrow 2\tau_*$, the tail factor tends to zero. If $u_n/(\ell_n - \tau_n) \rightarrow \infty$, the recent base can vanish even while the tail remains subunit. Diagonal examples realize all three boundaries. Thus the next asymptotic work is not an unspecified compactness argument: it must control overlap conditioning, the selected weak eigenvalue relative to the universal tail mass, and the selected spectral spread.

5 What has and has not closed

The finite native route is now internally complete: source-memory balls select reset packets, overlaps provide coherent charts, correlated congruence preserves the tail ratio, a delayed suffix removes finite birth conditioning, and Theorem 1 supplies one common positive support floor.

However, this native functional uses the recent memory compressed directly to the reset packet. The older directional construction used a projected cross action, a four-direction frame, wedge volume, leading singular value, and capacity. Equality or domination between those functionals has not been proved. Nor have we established all-level endpoint laws, Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

- Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.
- Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013.